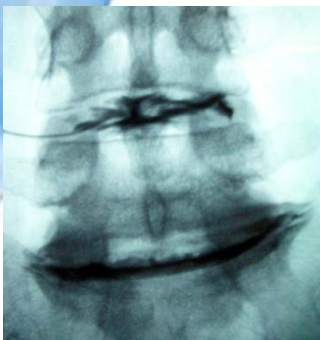


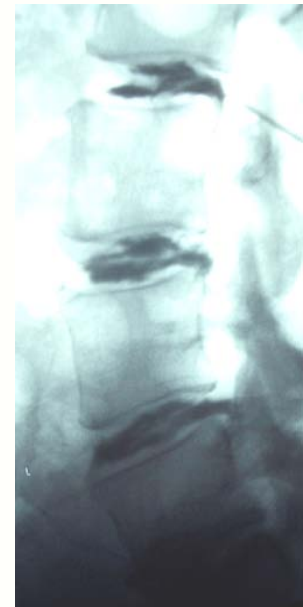
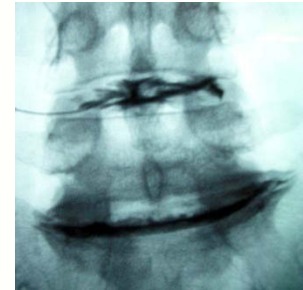
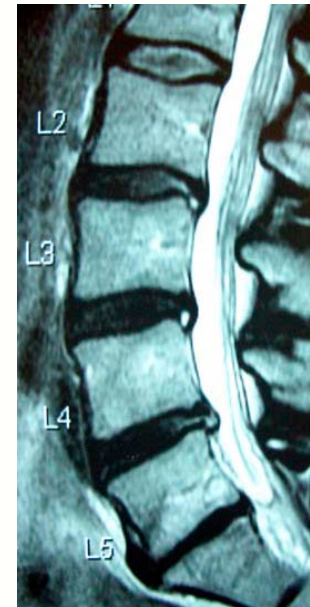
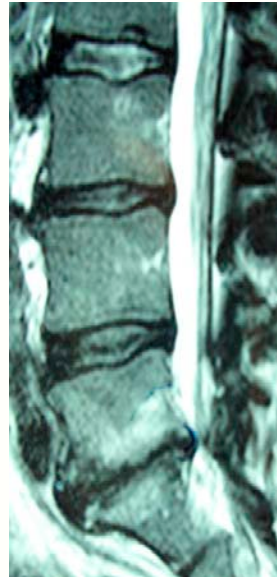
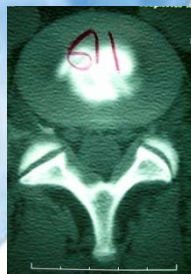
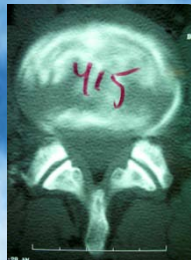
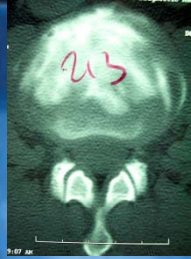
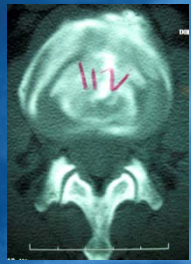
Discography



Pressure-monitored Discography
Concordant pain reproduction
Structural changes of the disc



Annular Tears and Imaging



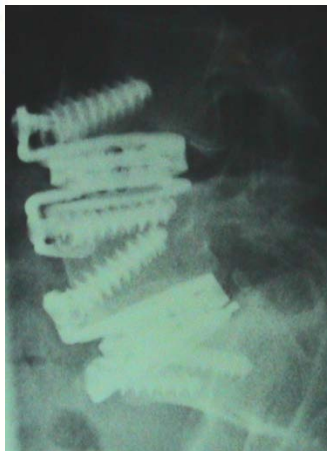
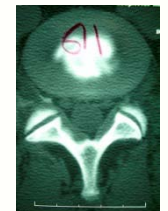
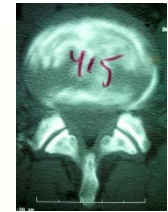
Location of Tears and Therapy



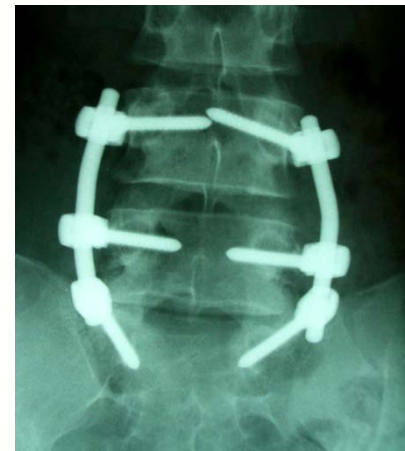
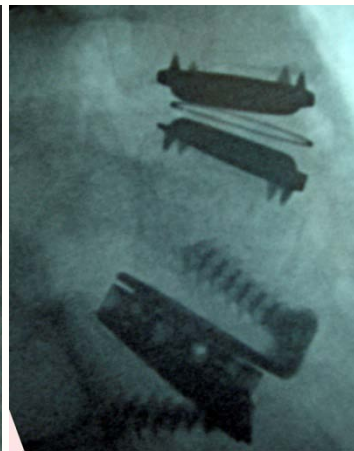
Anterior tear



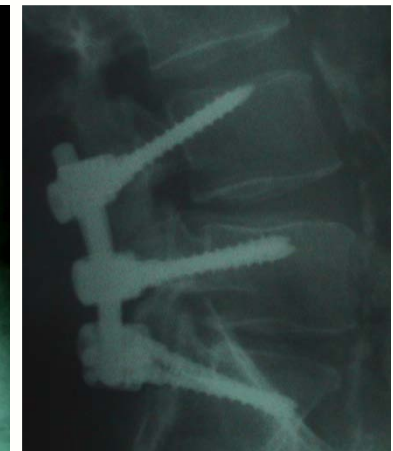
Posterior tear



Topaz/Charite

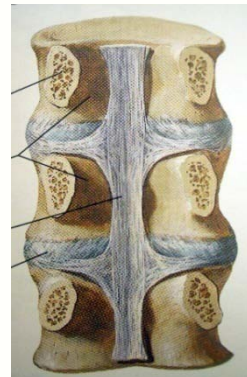
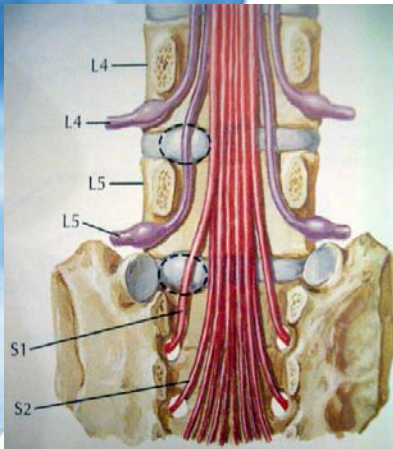


Cosmic



HLD (Herniated Lumbar Disc)

- **Anatomical background**
 - PLL strongest medially > HLD mostly posterolateral
- **Clinical presentation and Findings**
 - LBP +/- radicular pain
 - +/- sphincteric d/o (reduced bladder sensation)
 - (+)ve Valsalva-manoeuvres
 - (claudication) with central HLD
 - Senso-motor deficits >pix
 - SLR (L5-S1), FST (L2-L4), Trendelenburg (L5), FABER (hip)

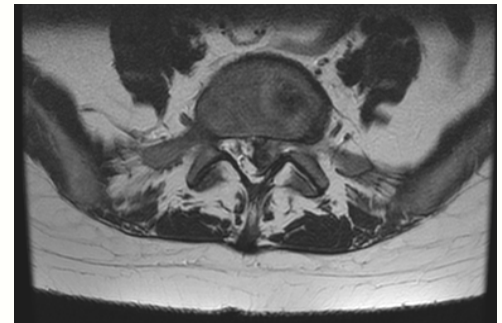


HLD (Herniated Lumbar Disc) Surgical Treatment

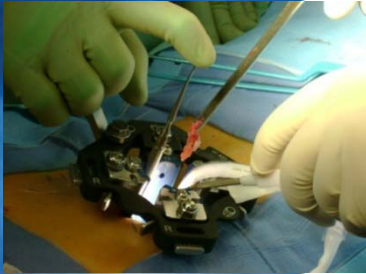
85% of acute HLD improve w/o surgery within 6/52

Surgical Indication:

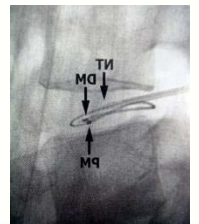
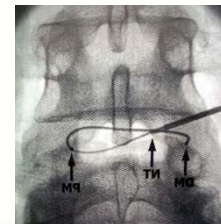
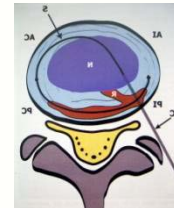
- Cauda Equina Syndrome (CES) 1-2% of all pts.
 - retention/incontinence/saddle anaesthesia/erectile d/o
 - outcome: only 50% regain function
- progressive motor deficit
- intractable pain



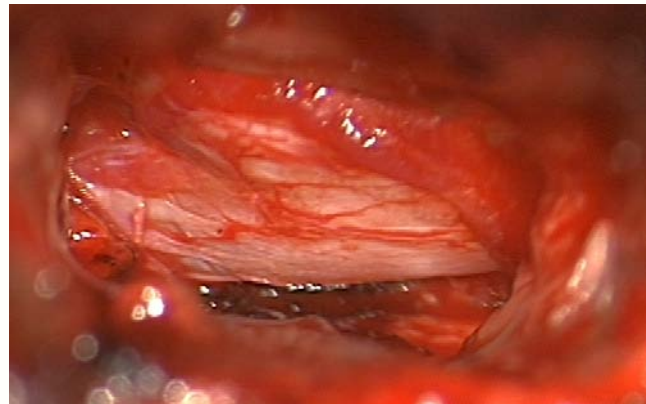
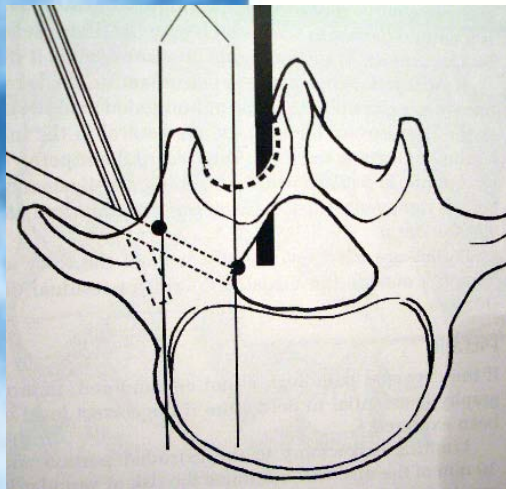
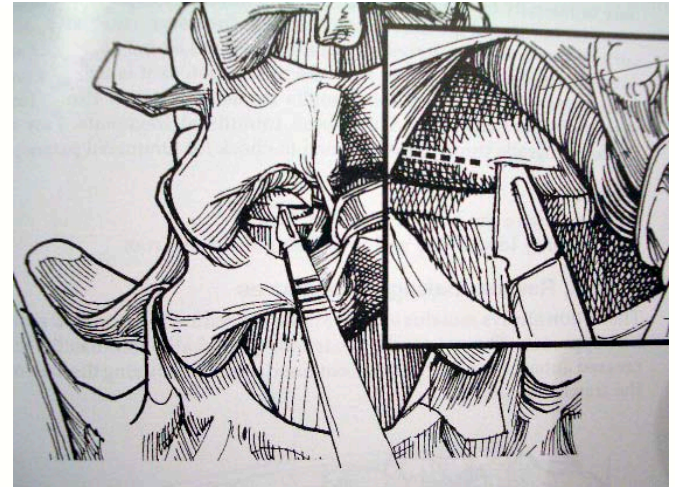
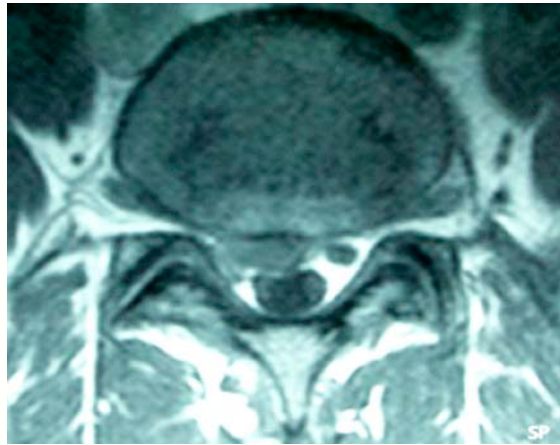
HLD (Herniated Lumbar Disc) Surgical Options



- (Open Discectomy/Laminectomy)
- Microdiscectomy (85% success rate @ 1yr.)
- Endoscopic Procedure
- Intradiscal Procedures
 - Chemonucleolysis (45% @ 1yr., anaphylaxis)
 - Nucleotomy (35% @ 1yr.)
 - IDET (40% @ 1yr.)



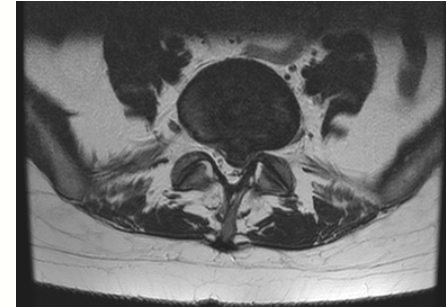
Microdiscectomy



HLD (Herniated Lumbar Disc) Complications

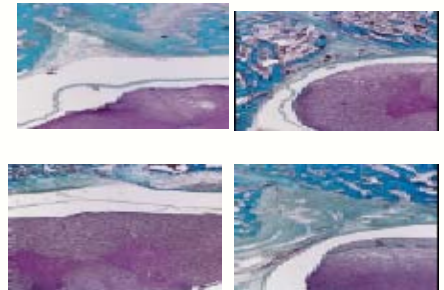
Common

- wound infection 0.5-1%
- motor deficit increased 1-8%
- dural tear CSF leak 0.5-13%
- scar formation (fat graft, barrier gels)

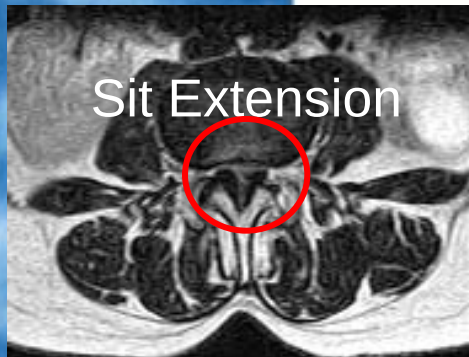
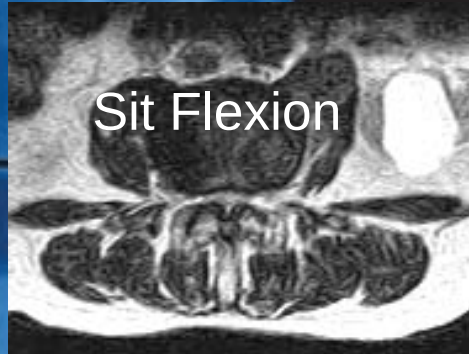


Uncommon

- anterior structure injury (vessels, ureter, bowel, sympathetic..)
- CES 0.2%
- arachnoiditis
- DVT with consecutive PE 0.1%



Spinal Stenosis Treatment



Nonsurgical Management

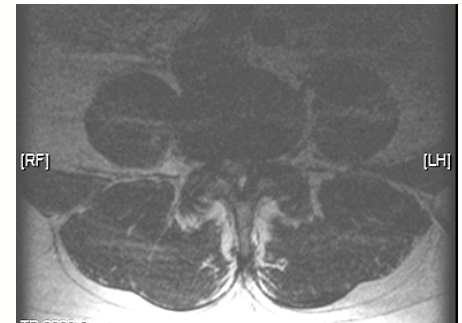
medication

PT +/- traction, general fitness and conditioning

injections (epidurals, DRGB)

TENS

LS-orthosis



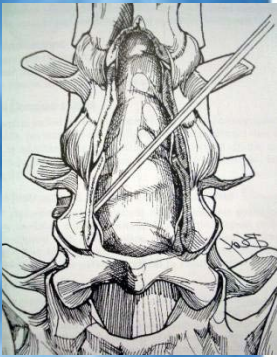
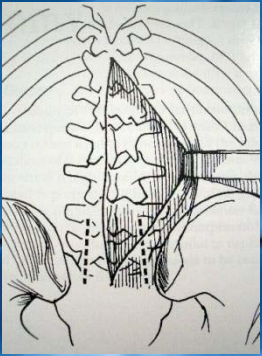
Surgical Treatment

is elective (Sx longer >3/12)

is emergent if impeding CES

Spinal Stenosis

Surgical Management



Goals

halting progression of Sx
pain relief
(reversal of existing Sx)

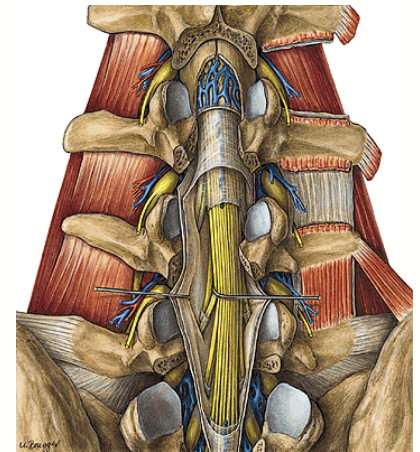
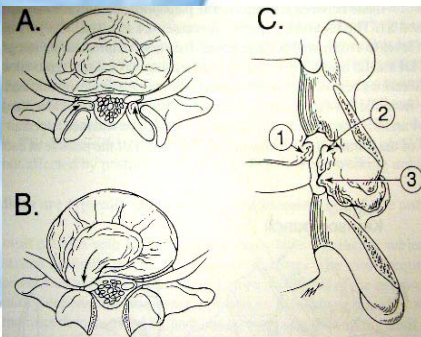


Options

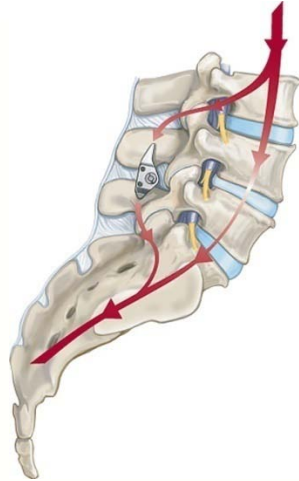
X-Stop (interspinous device)

Decompressive Procedures

(Fusion expected instability
excess facet removal >50%
pars Fx or removal
discectomy @ same time)



Spinal Stenosis Outcome



X-Stop

80% improve
no long-term data

Decompressive Procedures

short-term improvement very common
recurrence of Sx 30% of pts @ 5 yrs
long-term improvement 66%
complications: HLD, instability (0-1%)





ISD: RETAIN / FLEXUS

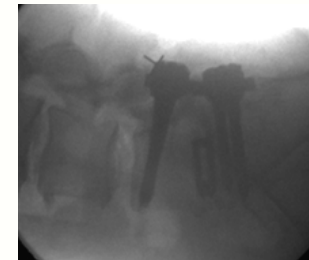
Outcome



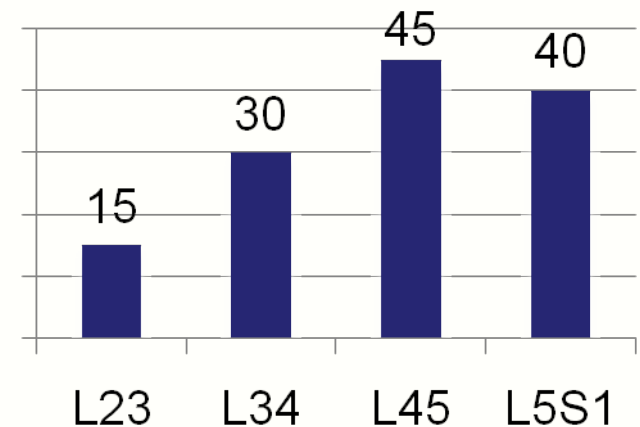
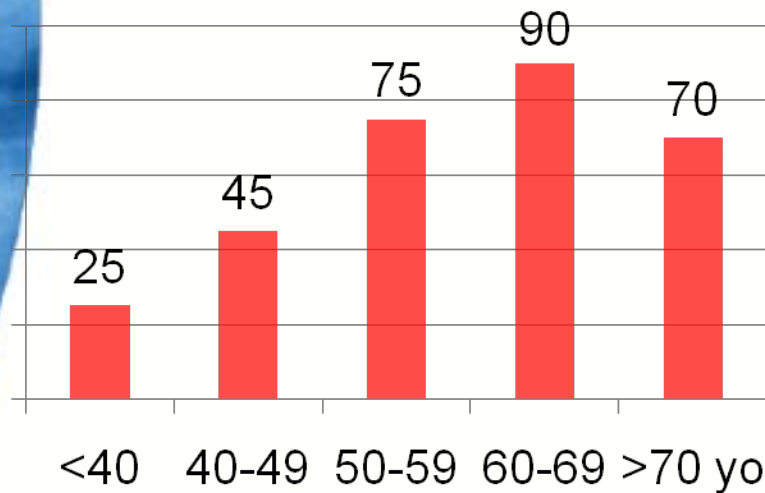
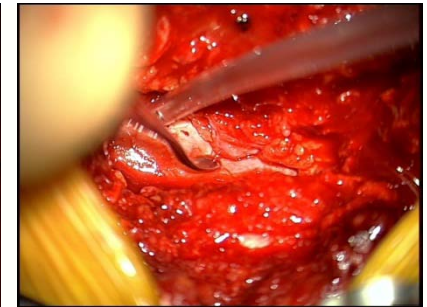
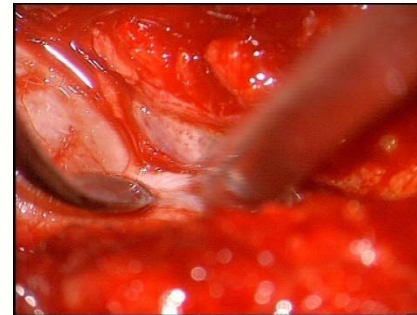
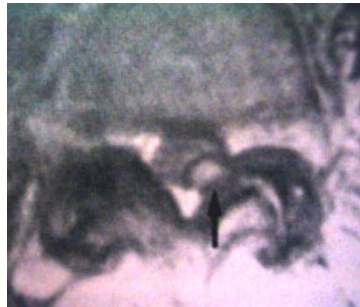
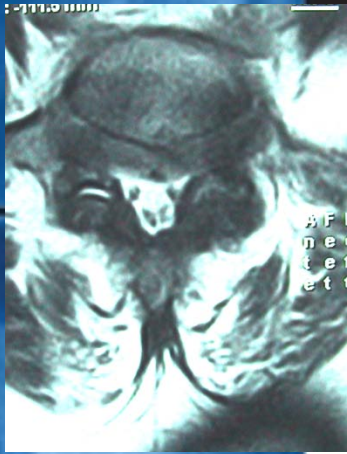
- Improved radicular pain 15 /21 pat.
- Improved radicular deficits (motor) 4 /6
- Improved LBP (FJ Cyst group) 4 /5
- Improved posture (subjective) 15 /21



- No change in c/o 3 /21 (14%)
- Overall improvement 18 /21 (85%)
- Minimally invasive
- Minimally disruptive
- Unilateral approach



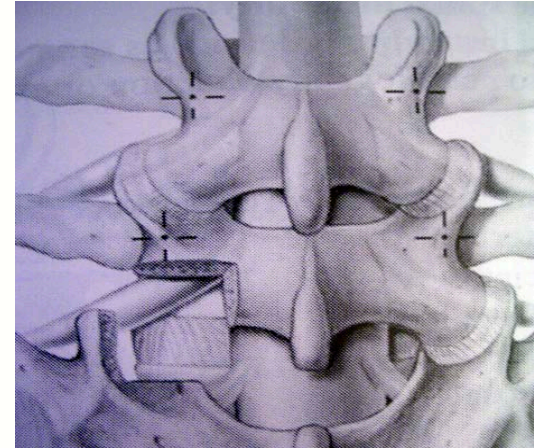
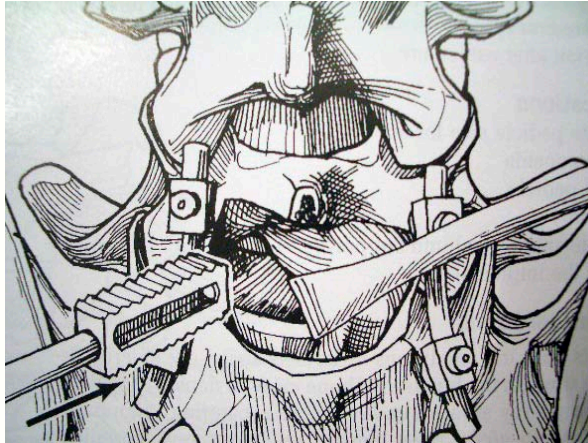
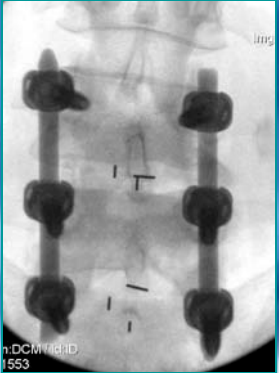
Facet Joint Osteoarthritis FJ OA



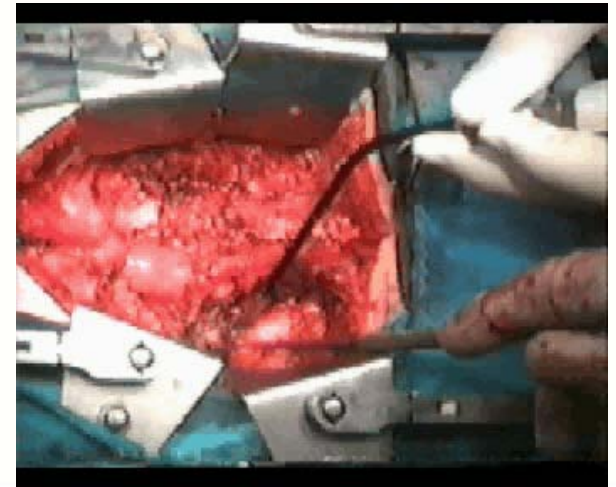
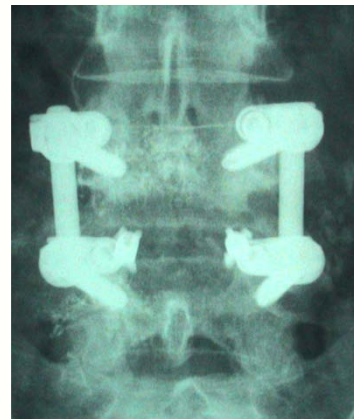
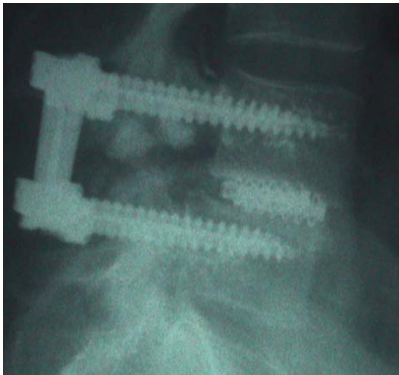
Facet Joint Osteoarthritis and Low Back Pain in the Community-Based Population.
Kalichman et al.: Spine Nov 2008

PLIF/ TLIF

(Posterior/ Transforaminal
Lumbar Interbody Fusion)



PLIF



Posterior Stabilization

- Problems/Complications
 - Anterior structures
 - Posterior/foraminal structures
 - Muscle irritation (profile of hardware)
 - Battery effect (different materials)

